

WEEKLY PROGRESS REPORT

Week 1 – 31.05.2026

Project Title: Implicit Domain Adaptation Through Pretrained Feature Reuse: An Empirical Study of Lightweight Transfer Learning Across Heterogeneous Medical Imaging Sources
Team Members: Abhay Sivakumar Nair (220905199) | Shruthi Andra (230905223)
Department: Computer Science & Engineering, MIT Manipal, MAHE
Guide: Dr. Tyson B D Cunha
Report Period: Week 1 | May 23 - May 30, 2026
Specialization: Minor Specialization in Computational Intelligence

1. Project Abstract

This project investigates whether lightweight ImageNet-pretrained convolutional neural networks (EfficientNetB0, MobileNetV2, ResNet50) provide implicit domain adaptation across heterogeneous medical imaging sources, eliminating the need for computationally expensive explicit alignment mechanisms such as adversarial domain adaptation or GAN-based style transfer.

The core hypothesis, termed implicit domain adaptation through pretrained feature reuse, is that pretrained architectures encode sufficiently domain-invariant feature representations to bridge scanner-induced and population-induced distribution shifts. This hypothesis is directly motivated by prior empirical work in structural crack detection, where EfficientNetB0 outperformed a custom CNN by approximately 10 percentage points on pooled multi-source data with characteristic convergence instability in the custom model.

The project validates this hypothesis across three medical imaging tasks: skin lesion classification (ISIC 2019, HAM10000), chest X-ray pathology detection (CheXpert, VinDr-CXR, NIH ChestX-ray14), and diabetic retinopathy grading (EyePACS, APTOS 2019, Messidor-2). Domain gap is quantified using Maximum Mean Discrepancy (MMD) and mechanistic interpretability is provided through Grad-CAM analysis.

2. Work Completed This Week

2.1 Dataset Acquisition and Pipeline Setup

Two skin lesion datasets were identified, verified, and integrated into a unified training pipeline on Kaggle:

Dataset	Source	Images	Classes	Role
ISIC 2019	Global (multi-institution)	25,331	6 shared	Domain 1
HAM10000	Vienna + Australia	9,688	6 shared	Domain 2
Combined	Multi-source pooled	35,019	6	Experiment dataset

Both datasets were filtered to retain only the six diagnostic classes shared across both sources: melanoma, nevus, basal cell carcinoma (bcc), benign keratosis (bkl), dermatofibroma (df), and vascular lesion (vasc). Images were tagged by source to enable domain-aware analysis. Final combined dataset: 35,019 images across 6 classes and 2 sources.

2.2 Data Pipeline Implementation

A complete data pipeline was implemented including:

- Stratified 70/15/15 train/val/test split applied independently per source to ensure proportional source representation across all splits
- Class weight computation using sklearn to address severe class imbalance (nevus: 19,580 images vs df: 354 images)
- Separate ImageDataGenerator configurations for custom CNN (rescale to 0-1) and EfficientNetB0 (raw 0-255 via efficientnet_preprocess) to ensure correct preprocessing for each architecture
- Data augmentation applied to training set: rotation, shifts, horizontal flip, zoom

2.3 Model Training: Custom CNN on Pooled Multi-Source Data

A 3-block custom CNN (322,470 parameters) was trained from scratch on the pooled ISIC 2019 + HAM10000 dataset for 27 epochs (early stopping, patience=7). Key observations:

- Best validation accuracy: 54.58% at epoch 20
- Training showed characteristic oscillation: val accuracy ranged from 11.14% (epoch 1) to 54.58% (epoch 20) with no stable convergence
- Training accuracy reached 52.39% while val accuracy oscillated, indicating difficulty generalizing across the two imaging sources
- This instability pattern directly replicates the convergence instability observed in the crack detection baseline, validating the experimental design

2.4 Model Training: EfficientNetB0 on Pooled Multi-Source Data

EfficientNetB0 (4,379,049 parameters, ImageNet pretrained) was trained in two phases:

- Phase 1 (frozen backbone, head only): 10 epochs, best val accuracy 65.56%, smooth monotonic improvement from epoch 1

2.5 Preliminary Results

Model	Val Accuracy (Best)	Epochs Run	Training Stability
Custom CNN (from scratch)	54.58%	27 (early stop)	Highly unstable, oscillating
EfficientNetB0 Phase 1 (frozen backbone)	65.56%	10	Smooth, consistent climb

The 10.98 percentage point gap between EfficientNetB0 (65.65%) and custom CNN (54.58%) on pooled multi-source data constitutes the core preliminary finding. The stability contrast between models is equally significant: EfficientNetB0 improved monotonically across all 10 epochs while the custom CNN oscillated throughout training.

2.6 Technical Infrastructure

- Platform: Kaggle Notebooks with Tesla T4 x2 GPU
- All model weights saved to /kaggle/working/ with CSVLogger for training history persistence

3. Plan for Next Week (June 1-7)

- Train EfficientNetB0 Phase 2 – unfreeze the backbone, run it across 40 epochs, note the best val accuracy 65.56% and smooth monotonic improvement from epoch 1

4. Issues and Risks

- Kaggle GPU quota: 30 hours per week free tier. Each full training run consumes approximately 14 hours. Quota management is an active constraint. Mitigation: training scheduled for overnight runs, Save Version used after each completed training block.
- BCN20000 dataset: No clean verified version available on Kaggle. Resolved by using ISIC 2019 and HAM10000 as two distinct sources. These represent genuinely different acquisition contexts (global multi-institution vs Vienna/Australia) sufficient for domain gap demonstration.
- PhysioNet credentialling for chest datasets: Approval timeline of 3-7 days. Will initiate this process in June.

6. Declaration

We declare that the work reported in this weekly report has been carried out by us as part of the Minor Specialization project in Computational Intelligence. All experiments were conducted on publicly available datasets under their respective licenses.

Abhay Sivakumar Nair
Reg. No: 220905199

Shruthi Andra
Reg. No: 230905223

Date: May 31, 2026

Date: May 31, 2026
